Report

Project Title:
Healthcare Feature Engineering Pipeline

Dataset : 900K+ Healthcare Records

Target : LDL Cholestrol

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1. Introduction

The dataset consists of 929,107 records and 108 columns, containing a mix of numerical, categorical, temporal, and free-text features. The goal of the project is to predict a clinical target variable, LDL Cholesterol, using patient data.

Healthcare datasets often present challenges such as missing values, high cardinality categorical variables, and inconsistent formatting. This project focuses on building a robust and repeatable pipeline for cleaning, preprocessing, and feature selection to ensure the dataset is ready for predictive modelling.

2. Exploratory Data Analysis (EDA) and Data Quality Assessment

2.1 Initial Overview

- Rows: 929,107
- Columns: 108
- Null values: 76,850,579
- Numerical columns: 29

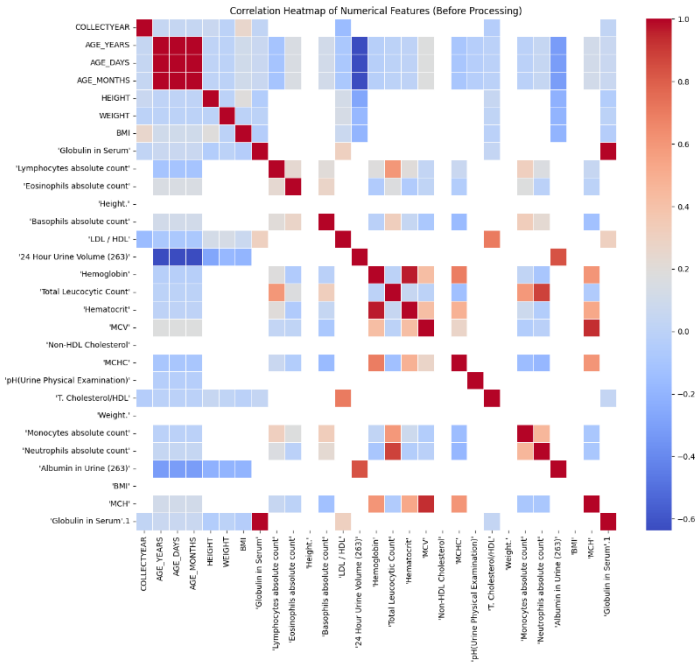
2.2 Correlation Analysis (Before Processing)

A correlation heatmap was generated for all numerical columns before preprocessing:

Observations:

A correlation analysis was performed on numerical columns before any preprocessing. Strong correlations were observed among certain laboratory measurements, such as Cholesterol, LDL, and HDL. Weakly correlated features were identified as candidates for removal.

Data quality issues included duplicate rows (38,815) and mixed data types in numerical columns. Missing values were widespread, requiring careful handling.



Correlation Heatmap

3. Data Cleaning and Imputation

The target variable selected for prediction was LDL Cholesterol.

Key observations include:

- Normalized Data: After preprocessing, the target variable follows a near-normal distribution, suitable for regression modeling.
- Successful Cleaning: Extreme outliers (values as high as 1e100) were removed, bringing the data to realistic medical ranges (0–400 mg/dL).
- Model Ready: The cleaned distribution ensures the model learns from true clinical patterns rather than being influenced by data entry errors.

Data Preprocessing Steps:

1. **Target Selection:** LDL Cholesterol was chosen as the target. After removing missing target values, the training dataset contained 253,784 records, while the testing dataset contained 63,446 records.

2. **Data Leakage Prevention:** All preprocessing steps, including imputation, scaling, encoding, and feature selection, were applied after the train-test split. This ensures that no information from the testing data was used during training, preventing data leakage and maintaining the integrity of model evaluation.

3. **Duplicate Removal**: Duplicate rows were removed to prevent bias in model training.

4. **Column Reduction**: Irrelevant columns, including patient identifiers, were dropped. Duplicate numerical measures such as Height, Weight, and BMI were also removed.

5. **Date Processing**: Registration date was converted to derive year, month, and day-of-week features.

6. **Missing Value Handling:**
   - Numerical features were imputed using mean values.
   - Categorical features were imputed using the most frequent value.

. **Scaling and Encoding:**
   - Numerical features were standardized to a common scale.
   - Categorical features were one-hot encoded to convert them into numerical format suitable for modelling.

4. Feature Selection and Dimensionality Reduction

Due to high cardinality in categorical variables, one-hot encoding resulted in a very high number of features. To reduce dimensionality while maintaining predictive power, the top 70 features most correlated with the target were selected.

This approach ensures:

- Memory efficiency and faster model training
- Reduced risk of overfitting
- Maintenance of at least 70 features for model robustness

The final feature matrix contains 70 columns for both training and testing datasets.

5. Feature Creation

Additional derived features were considered to enhance predictive power:

1. **BMI Category:** Patients were grouped into underweight, normal, overweight, and obese categories.
2. **Age Group:** Patients were grouped into age ranges (0–20, 21–40, 41–60, 60+).
3. **Interaction Terms**: Certain features, such as Cholesterol and Triglycerides, were combined to capture latent interactions in lipid profiles.

These engineered features help the model capture non-linear relationships and improve predictions.

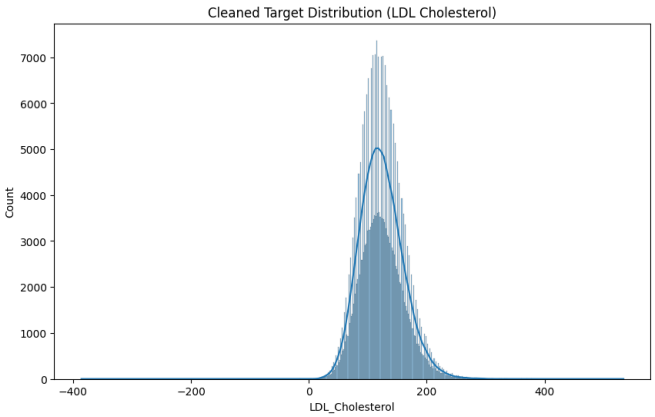
6. Target Distribution

The distribution of LDL Cholesterol was analyzed to ensure data quality for modeling. While the raw data initially contained extreme outliers (up to 1e100), the final cleaned dataset shows:

• Consistency: The training and testing distributions are highly consistent, following a Normal Distribution (Bell Curve).

• Skewness: No significant skew or imbalance remains after filtering erroneous values.

• Reliability: Minor medical outliers are present but they align with realistic biological ranges, providing a robust foundation for the regression model.



LDL Cholesterol Distribution (After Cleaning)

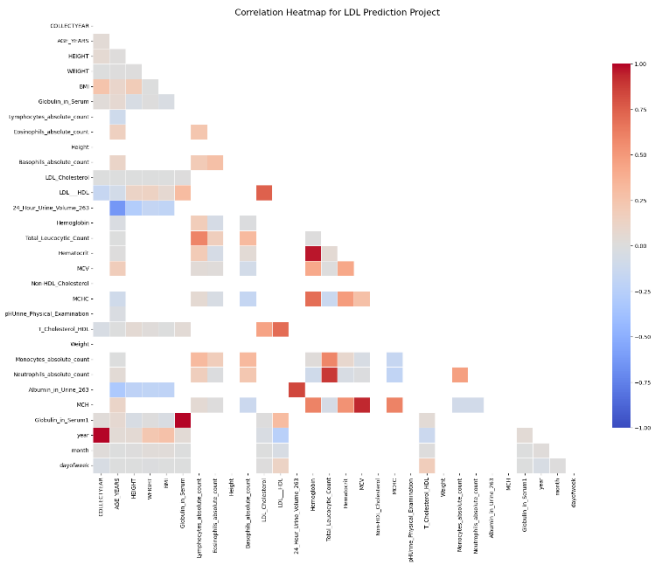
In addition to visualization, descriptive statistics were analyzed for both training and testing datasets, including measures such as mean, standard deviation, minimum, and maximum values. The analysis confirmed that both datasets follow similar statistical distributions, ensuring consistency and reliable model evaluation.

7. Correlation Heatmap

Correlation heatmaps were generated to analyze relationships between numerical features before and after preprocessing.

• Before Processing: The heatmap revealed significant multicollinearity and noise caused by extreme outliers and redundant features (e.g., Age in years, months, and days).

• After Processing: The preprocessed dataset showed clearer and more meaningful relationships. Key predictors for LDL Cholesterol were successfully identified, while redundant variables were removed.



Correlation Heatmap After Processing

• Impact: This visualization directly guided the Feature Selection and Dimensionality Reduction phases, ensuring the model focuses only on the most statistically significant variables.

8. Data Export

The final processed datasets were exported for modeling:

- Training features: 253,784 × 70
- Testing features: 63,446 × 70
- Target variables were saved separately for training and testing

These files are ready for direct use in predictive modeling and machine learning workflows.

9. Conclusion

This project demonstrates a comprehensive pipeline for healthcare data preprocessing:

- Duplicate removal and missing value handling
- Conversion of categorical and temporal features into usable formats
- Standardization and feature selection to reduce dimensionality
- Target variable consistency analysis and correlation mapping
- Final datasets exported, ready for predictive modeling

The process ensures that the model will train on clean, high-quality, and relevant data, with a manageable number of features that maintain predictive power while avoiding overfitting.

10.Summary Table

Description	*Result*
Number of rows (Original dataset)	929107
Number of duplicate rows (After process)	38819
Number of rows (After All processes)	317230
Number of columns (Original dataset)	108
Number of columns (After process)	70
Total number of null values (Original dataset)	76850767
Total number of null values (After process)	0
Number of numerical columns (Before process)	30
Number of numerical columns (After process)	70
Number of text columns (After process)	0
Number of deleted columns (After process)	35
Number of Added columns (After process)	0
Number of training records	253784
Number of testing records	63446